RAJALAKSHMI ENGINEERING COLLEGE

Harshini J 250701233-2025-CSE

H2

Started on	Tuesday, 6 October 2026, 2:24 PM
State	Finished
Completed on	Sunday, 11 October 2026, 8:24 AM
Time taken	4 days 17 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, i, x;
    int *seen;
    int found = 0;

    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("-1\n");
        return 0;
    }

    seen = (int *)calloc(n + 1, sizeof(int));
    if (seen == NULL)
        return 1;

    for (i = 0; i < n; i++) {
        if (scanf("%d", &x) != 1)
            break;
        if (!found && x >= 1 && x <= n) {
            if (seen[x]) {
                printf("%d\n", x);
                found = 1;
            } else {
                seen[x] = 1;
            }
        }
    }

    if (!found)
        printf("-1\n");

    free(seen);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Harshini J 250701233-2025-CSE

H2

Started on	Tuesday, 6 October 2026, 2:26 PM
State	Finished
Completed on	Sunday, 11 October 2026, 8:25 AM
Time taken	4 days 17 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, i;
    int *a, *count;

    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("-1\n");
        return 0;
    }

    a = (int *)malloc(n * sizeof(int));
    count = (int *)calloc(n + 1, sizeof(int));
    if (a == NULL || count == NULL)
        return 1;

    for (i = 0; i < n; i++) {
        if (scanf("%d", &a[i]) != 1)
            a[i] = 0;
        if (a[i] >= 1 && a[i] <= n)
            count[a[i]]++;
    }
    for (i = 0; i < n; i++) {
        if (a[i] >= 1 && a[i] <= n && count[a[i]] > 1) {
            printf("%d\n", a[i]);
            free(a);
            free(count);
            return 0;
        }
    }

    printf("-1\n");
    free(a);
    free(count);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Started on	Friday, 9 October 2026, 11:39 PM
State	Finished
Completed on	Friday, 9 October 2026, 11:40 PM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2, i = 0, j = 0;
        scanf("%d", &n1);

        int a[n1];
        for (int k = 0; k < n1; k++)
            scanf("%d", &a[k]);

        scanf("%d", &n2);

        int b[n2];
        for (int k = 0; k < n2; k++)
            scanf("%d", &b[k]);

        int first = 1;
```

```
        int first = 1;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                if (!first)
                    printf(" ");
                printf("%d", a[i]);
                first = 0;
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RAJALAKSHMI ENGINEERING COLLEGE

Harshini J 250701233-2025-CSE

H2

Started on	Wednesday, 7 October 2026, 10:13 PM
State	Finished
Completed on	Sunday, 11 October 2026, 8:25 AM
Time taken	3 days 10 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int t;
    scanf("%d", &t);
    while (t--) {
        int n1, n2;

        scanf("%d", &n1);
        int *a = (int *)malloc(n1 * sizeof(int));
        for (int k = 0; k < n1; k++)
            scanf("%d", &a[k]);

```

```
        scanf("%d", &n2);
        int *b = (int *)malloc(n2 * sizeof(int));
        for (int k = 0; k < n2; k++)
            scanf("%d", &b[k]);

        int i = 0, j = 0, first = 1;
        while (i < n1 && j < n2) {
            if (a[i] < b[j]) {
                i++;
            } else if (a[i] > b[j]) {
                j++;
            } else {
                if (!first) printf(" ");
                printf("%d", a[i]);
                first = 0;
                i++;
                j++;
            }
        }
        printf("\n");

        free(a);
        free(b);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Harshini J 250701233-2025-CSE

H2

Started on	Wednesday, 7 October 2026, 10:14 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:34 PM
Time taken	20 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, k;
    scanf("%d", &n);
    int *a = (int *)malloc(n * sizeof(int));
    for (int x = 0; x < n; x++)
        scanf("%d", &a[x]);
    scanf("%d", &k);

    int i = 0, j = 1, found = 0;
    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }
        int diff = a[j] - a[i];
        if (diff == k) {
            found = 1;
            break;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }
    }

    printf("%d\n", found);
    free(a);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Harshini J 250701233-2025-CSE

H2

Started on	Wednesday, 7 October 2026, 10:16 PM
State	Finished
Completed on	Sunday, 11 October 2026, 8:26 AM
Time taken	3 days 10 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    long long k;
    scanf("%d", &n);
    long long *a = (long long *)malloc(n * sizeof(long long));
    for (int x = 0; x < n; x++)
        scanf("%lld", &a[x]);
    scanf("%lld", &k);

    int i = 0, j = 1, found = 0;
    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }
        long long diff = a[j] - a[i];
        if (diff == k) {
            found = 1;
            break;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }
    }

    printf("%d\n", found);
    free(a);
    return 0;
```

```
33 }
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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